2501.06662 — formula report

386 inline formulas (section omitted; all rendered), 60 display equations, 0 tables, 0 unrecovered image regions.

Display equations

<table>
<tr><th>Identifier</th><th>Page</th><th>Conf.</th><th>LaTeX source</th><th>Rendered</th><th>Scan image</th></tr>
<tr><td>2501.06662_EQ0051</td><td>021</td><td>0.536
C|+4</td><td><code>\begin{aligned} \operatorname{Mag}(t \mathcal{M}) & =\sum_{x, y \in \mathrm{ob}(\mathcal{M})} \zeta_{t}^{-1}(x, y) \\ & =\sum_{x, y \in \mathrm{ob}(\mathcal{M})} \sum_{k \geq 0} \sum_{\substack{\text { nondeg.paths } \\ x=y_{0} \rightarrow y_{1} \rightarrow \cdots \rightarrow y_{k}=y}}(-1)^{k} \prod_{i=1}^{k} \pi \left(y_{i} \mid y_{i-1}\right)^{t} . \end{aligned}</code></td><td>$\begin{aligned} \operatorname{Mag}(t \mathcal{M}) & =\sum_{x, y \in \mathrm{ob}(\mathcal{M})} \zeta_{t}^{-1}(x, y) \\ & =\sum_{x, y \in \mathrm{ob}(\mathcal{M})} \sum_{k \geq 0} \sum_{\substack{\text { nondeg.paths } \\ x=y_{0} \rightarrow y_{1} \rightarrow \cdots \rightarrow y_{k}=y}}(-1)^{k} \prod_{i=1}^{k} \pi \left(y_{i} \mid y_{i-1}\right)^{t} . \end{aligned}$</td><td>$\begin{aligned} \operatorname{Mag}(t\mathcal{M}) & = \sum_{x,y\in\mathrm{ob}(\mathcal{M})} \zeta_t^{-1}(x,y) \\ & = \sum_{x,y\in\mathrm{ob}(\mathcal{M})} \sum_{k\geq 0} \sum_{\substack{\text{nondeg. paths} \\ x=y_0\to y_1\to\cdots\to y_k=y}} {}_{\text{in } \mathsf{L}} (-1)^k \prod_{i=1}^{k} \pi(y_i|y_{i-1})^t. \end{aligned}$</td></tr>
<tr><td>2501.06662_EQ0059</td><td>023</td><td>0.550
C|+5</td><td><code>\begin{aligned} \lim _{n \rightarrow \infty} & \frac{ \mathscr{H}\left(\mathcal{M},\left.\pi(-\mid \perp) \right|_{T(\perp)}, \zeta\right)}{n} \\ & =\lim _{n \rightarrow \infty}-\frac{1}{n} \sum_{\left(a_{1}, \ldots, a_{N-1}\right) \in A^{N-1}} \pi\left(\perp a_{1} \cdots a_{N-1} \mid \perp\right) \log \pi\left(\perp a_{1} \cdots a_{N-1} \mid \perp\right) \end{aligned}</code></td><td>$\begin{aligned} \lim _{n \rightarrow \infty} & \frac{ \mathscr{H}\left(\mathcal{M},\left.\pi(-\mid \perp) \right|_{T(\perp)}, \zeta\right)}{n} \\ & =\lim _{n \rightarrow \infty}-\frac{1}{n} \sum_{\left(a_{1}, \ldots, a_{N-1}\right) \in A^{N-1}} \pi\left(\perp a_{1} \cdots a_{N-1} \mid \perp\right) \log \pi\left(\perp a_{1} \cdots a_{N-1} \mid \perp\right) \end{aligned}$</td><td>$\begin{aligned} \lim_{n\to\infty} & \frac{\mathscr{H}(\mathcal{M}, \pi(-|\perp)|_{T(\perp)}, \zeta)}{n} \\ & = \lim_{n\to\infty} -\frac{1}{n} \sum_{(a_1,\ldots,a_{N-1})\in A^{N-1}} \pi(\perp a_1\cdots a_{N-1}|\perp) \log \pi(\perp a_1 \cdots a_{N-1}|\perp) \end{aligned}$</td></tr>
<tr><td>2501.06662_EQ0058</td><td>023</td><td>0.594
W|+1</td><td><code>\begin{aligned} \mathscr{H}(\mathcal{M}, p, \zeta) & =-\sum_{y \in \mathrm{ob}(\mathcal{M})} p(y) \log \left(\sum_{z \in \mathrm{ob}(\mathcal{M})} \zeta_{t}(y, z) p(z)\right) \\ & =-\sum_{y \in \mathrm{ob}(\mathcal{M})} p(y) \log \left(\sum_{z \in \mathrm{ob}(\mathcal{M})} \pi(z \mid y)^{t} p(z)\right) . \end{aligned}</code></td><td>$\begin{aligned} \mathscr{H}(\mathcal{M}, p, \zeta) & =-\sum_{y \in \mathrm{ob}(\mathcal{M})} p(y) \log \left(\sum_{z \in \mathrm{ob}(\mathcal{M})} \zeta_{t}(y, z) p(z)\right) \\ & =-\sum_{y \in \mathrm{ob}(\mathcal{M})} p(y) \log \left(\sum_{z \in \mathrm{ob}(\mathcal{M})} \pi(z \mid y)^{t} p(z)\right) . \end{aligned}$</td><td>$\begin{aligned} \mathscr{H}(\mathcal{M}, p, \zeta) & = -\sum_{y\in\mathrm{ob}(\mathcal{M})} p(y) \log \left(\sum_{z\in\mathrm{ob}(\mathcal{M})} \zeta_t(y,z)p(z) \right) \\ & = -\sum_{y\in\mathrm{ob}(\mathcal{M})} p(y) \log \left(\sum_{z\in\mathrm{ob}(\mathcal{M})} \pi(z|y)^t p(z) \right). \end{aligned}$</td></tr>
<tr><td>2501.06662_EQ0023</td><td>015</td><td>0.636
C|+8</td><td><code>\begin{aligned} \#\left\{x \in \mathrm{ob}(\mathrm{~L}): x=\perp a \text { with } a \in \bigcup_{i=0}^{N-2} A^{i} \right\} & (\# A+1)=\left(\sum_{i=0}^{N-2} \# A^{i} \right)(\# A+1) \\ & =\# A^{N-1}+2\left(\# A^{N-2} \right)+\cdots+2(\# A)+1 \end{aligned}</code></td><td>$\begin{aligned} \#\left\{x \in \mathrm{ob}(\mathrm{~L}): x=\perp a \text { with } a \in \bigcup_{i=0}^{N-2} A^{i} \right\} & (\# A+1)=\left(\sum_{i=0}^{N-2} \# A^{i} \right)(\# A+1) \\ & =\# A^{N-1}+2\left(\# A^{N-2} \right)+\cdots+2(\# A)+1 \end{aligned}$</td><td>$\begin{aligned} \# \left\{ x\in\mathrm{ob}(\mathsf{L}) : x = \perp a \text{ with } a \in \bigcup_{i=0}^{N-2} A^i \right\} (\#A+1) & = \left(\sum_{i=0}^{N-2} \#A^i \right) (\#A+1) \\ & = \#A^{N-1} + 2(\#A^{N-2}) + \cdots + 2(\#A) + 1. \end{aligned}$</td></tr>
</table>

Conf. MathPix confidence: ≥0.9 0.5–0.9 <0.5 — no value
Residual render vs scan (inkdrill): C component W weak S stable N noise K clean not measured

<table>
<tr><th>Identifier</th><th>Page</th><th>Conf.</th><th>LaTeX source</th><th>Rendered</th><th>Scan image</th></tr>
<tr><td>2501.06662_EQ0052 (13)</td><td>021</td><td>■ 0.693
● N|+1</td><td><code>\operatorname{Mag}(t \mathcal{M})=\sum_{k \geq 0} \sum_{\ell \in[0, \infty)} \sum_{c \in G_{k, \ell}}(-1)^{k} e^{-t \ell} .</code></td><td>$\operatorname{Mag}(t \mathcal{M})=\sum_{k \geq 0} \sum_{\ell \in[0, \infty)} \sum_{c \in G_{k, \ell}}(-1)^{k} e^{-t \ell}.$</td><td>$\operatorname{Mag}(t \mathcal{M})=\sum_{k \geq 0} \sum_{\ell \in[0, \infty)} \sum_{c \in G_{k, \ell}}(-1)^{k} e^{-t \ell}.$</td></tr>
<tr><td>2501.06662_EQ0033</td><td>017</td><td>■ 0.720
● W|+0</td><td><code>\zeta_{t}^{-1}(x, y)= \begin{cases}-\pi(y \mid x)^{t} & \text { if } y \in \mathrm{~L}_{x}^{(1)} \\ 1 & \text { if } y=x \\ 0 & \text { otherwise }\end{cases}</code></td><td>$\zeta_{t}^{-1}(x, y)= \begin{cases}-\pi(y \mid x)^{t} & \text { if } y \in \mathrm{~L}_{x}^{(1)} \\ 1 & \text { if } y=x \\ 0 & \text { otherwise }\end{cases}$</td><td>$\zeta_{t}^{-1}(x, y)= \begin{cases}-\pi(y \mid x)^{t} & \textit{if } y \in \mathsf{L}_{x}^{(1)} \\ 1 & \textit{if } y=x \\ 0 & \textit{otherwise.}\end{cases}$</td></tr>
<tr><td>2501.06662_EQ0037 (10)</td><td>018</td><td>■ 0.780
● W|+1</td><td><code>\begin{aligned} \operatorname{Mag}(t \mathcal{M}) & =\sum_{x, y \in \mathrm{ob}(\mathcal{M})} \zeta_{t}^{-1}(x, y) \\ & =\sum_{x \in \mathrm{ob}(\mathcal{M})} \zeta_{t}^{-1}(x, x)-\sum_{x \in \mathrm{ob}(\mathcal{M}) \backslash T(\perp)} \sum_{y \in \mathrm{~L}_{x}^{(1)}} \pi(y \mid x)^{t} \\ & =\sum_{x \in \mathrm{ob}(\mathcal{M}) \backslash T(\perp)}\left(1-\sum_{a \in A \cup\{\dagger\}} p_{x}(a)^{t}\right)+\#(T(\perp)) . \end{aligned}</code></td><td>$\begin{aligned} \operatorname{Mag}(t \mathcal{M}) & =\sum_{x, y \in \mathrm{ob}(\mathcal{M})} \zeta_{t}^{-1}(x, y) \\ & =\sum_{x \in \mathrm{ob}(\mathcal{M})} \zeta_{t}^{-1}(x, x)-\sum_{x \in \mathrm{ob}(\mathcal{M}) \backslash T(\perp)} \sum_{y \in \mathrm{~L}_{x}^{(1)}} \pi(y \mid x)^{t} \\ & =\sum_{x \in \mathrm{ob}(\mathcal{M}) \backslash T(\perp)}\left(1-\sum_{a \in A \cup\{\dagger\}} p_{x}(a)^{t}\right)+\#(T(\perp)) . \end{aligned}$</td><td>$\begin{aligned} \operatorname{Mag}(t \mathcal{M}) & =\sum_{x, y \in \mathrm{ob}(\mathcal{M})} \zeta_{t}^{-1}(x, y) \\ & =\sum_{x \in \mathrm{ob}(\mathcal{M})} \zeta_{t}^{-1}(x, x)-\sum_{x \in \mathrm{ob}(\mathcal{M}) \backslash T(\perp)} \sum_{y \in \mathsf{L}_{x}^{(1)}} \pi(y \mid x)^{t} \\ & =\sum_{x \in \mathrm{ob}(\mathcal{M}) \backslash T(\perp)}\left(1-\sum_{a \in A \cup\{\dagger\}} p_{x}(a)^{t}\right)+\#(T(\perp)). \end{aligned}$</td></tr>
<tr><td>2501.06662_EQ0049</td><td>020</td><td>■ 0.797
● C|+3</td><td><code>\begin{aligned} & \partial_{k}^{i}\left(y_{0}, \ldots, y_{k}\right)= \\ & \begin{cases}\left(y_{0}, \ldots, y_{i-1}, y_{i+1}, \ldots, y_{k}\right) & \text { if } d\left(y_{i-1}, y_{i}\right)+d\left(y_{i}, y_{i+1}\right)=d\left(y_{i-1}, y_{i+1}\right) \\ 0 & \text { otherwise }\end{cases} \end{aligned}</code></td><td>$\begin{aligned} & \partial_{k}^{i}\left(y_{0}, \ldots, y_{k}\right)= \\ & \begin{cases}\left(y_{0}, \ldots, y_{i-1}, y_{i+1}, \ldots, y_{k}\right) & \text { if } d\left(y_{i-1}, y_{i}\right)+d\left(y_{i}, y_{i+1}\right)=d\left(y_{i-1}, y_{i+1}\right) \\ 0 & \text { otherwise }\end{cases} \end{aligned}$</td><td>$\begin{aligned} & \partial_{k}^{i}\left(y_{0}, \ldots, y_{k}\right)= \\ & \begin{cases}\left(y_{0}, \ldots, y_{i-1}, y_{i+1}, \ldots, y_{k}\right) & \text { if } d\left(y_{i-1}, y_{i}\right)+d\left(y_{i}, y_{i+1}\right)=d\left(y_{i-1}, y_{i+1}\right) \\ 0 & \text { otherwise }\end{cases} \end{aligned}$</td></tr>
<tr><td>2501.06662_EQ0001</td><td>003</td><td>■ 0.822
● N|+0</td><td><code>\operatorname{Mag}(t \mathcal{M})=(t-1) \sum_{x \in \operatorname{ob}(\mathcal{M}) \backslash T(\perp)} H_{t}\left(p_{x}\right)+\#(T(\perp)), \quad t>0 .</code></td><td>$\operatorname{Mag}(t \mathcal{M})=(t-1) \sum_{x \in \operatorname{ob}(\mathcal{M}) \backslash T(\perp)} H_{t}\left(p_{x}\right)+\#(T(\perp)), \quad t>0 .$</td><td>$\operatorname{Mag}(t \mathcal{M})=(t-1) \sum_{x \in \operatorname{ob}(\mathcal{M}) \backslash T(\perp)} H_{t}(p_{x})+\#(T(\perp)), \qquad t>0.$</td></tr>
<tr><td>2501.06662_EQ0029</td><td>016</td><td>■ 0.829
● W|+1</td><td><code>\left(\zeta_{t}-\delta\right)^{k}(x, y)=\sum_{\substack{\left(y_{0}, y_{1}, \ldots, y_{k-1}, y_{k}\right) \\ y_{0}=x, y_{k}=y \\ \left(y_{i-1}, y_{i}\right) \in E \text { for } i=1, \ldots, k}} \prod_{i=1}^{k} \pi \left(y_{i} \mid y_{i-1}\right)^{t}</code></td><td>$\left(\zeta_{t}-\delta\right)^{k}(x, y)=\sum_{\substack{\left(y_{0}, y_{1}, \ldots, y_{k-1}, y_{k}\right) \\ y_{0}=x, y_{k}=y \\ \left(y_{i-1}, y_{i}\right) \in E \text { for } i=1, \ldots, k}} \prod_{i=1}^{k} \pi \left(y_{i} \mid y_{i-1}\right)^{t}$</td><td>$(\zeta_{t}-\delta)^{k}(x, y)=\sum_{\substack{(y_{0}, y_{1}, \ldots, y_{k-1}, y_{k}) \\ y_{0}=x, y_{k}=y \\ (y_{i-1}, y_{i}) \in E \text { for } i=1, \ldots, k}} \prod_{i=1}^{k} \pi(y_{i} \mid y_{i-1})^{t}.$</td></tr>
</table>

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
2501.06662_ EQ0036 (11)	018	■ 0.835 ● N|+0	\operatorname{Mag}(t \mathcal{M})=(t-1) \sum_{x \in \operatorname{ob}(\mathcal{M}) \backslash T(\perp)} H_{t} \left(p_{x}\right)+\#(T(\perp)), \quad t>0 .	$\operatorname{Mag}(t \mathcal{M})=(t-1) \sum_{x \in \operatorname{ob}(\mathcal{M}) \backslash T(\perp)} H_{t} \left(p_{x}\right)+\#(T(\perp)), \quad t>0 .$	$\operatorname{Mag}(t\mathcal{M})=(t-1)\sum_{x\in\operatorname{ob}(\mathcal{M})\setminus T(\perp)} H_t(p_x)+\#(T(\perp)), \qquad t>0.$
2501.06662_ EQ0026 (8)	015	■ 0.842 ● C|+4	\zeta_{t}^{-1}(x, y)=\sum_{k \geq 0} \sum_{\substack{ \text { nondeg.paths } \\ x=y_{0} \rightarrow y_{1} \rightarrow \cdots \rightarrow y_{k}=y}}(-1)^{k} \prod_{i=1}^{k} \pi\left(y_{i} \mid y_{i-1}\right)^{t} .	$\zeta_{t}^{-1}(x, y)=\sum_{k \geq 0} \sum_{\substack{ \text { nondeg.paths } \\ x=y_{0} \rightarrow y_{1} \rightarrow \cdots \rightarrow y_{k}=y}}(-1)^{k} \prod_{i=1}^{k} \pi\left(y_{i} \mid y_{i-1}\right)^{t} .$	$\zeta_t^{-1}(x,y)=\sum_{k\geq 0}\sum_{\substack{\textit{nondeg. paths}\\ x=y_0\to y_1\to\cdots\to y_k=y}} \textit{in } \mathsf{L} \quad (-1)^k\prod_{i=1}^{k}\pi(y_i\|y_{i-1})^t.$
2501.06662_EQ0021	014	■ 0.849 ● N|+2	\begin{aligned} \operatorname{Mag}(\mathrm{L}) & = \sum_{x, y \in \mathrm{ob}(\mathrm{~L})} \zeta_{ \mathrm{L}}^{-1}(x, y) \\ & =\sum_{x, y \in \mathrm{ob}(\mathrm{~L})} \mu_{\mathrm{L}}(x, y) \\ & =\sum_{x \in \mathrm{ob}(\mathrm{~L})} \mu_{\mathrm{L}}(x, x)+ \sum_{\substack{x=\perp a \in \mathrm{ob}(\mathrm{~L}) \\ a \in \bigcup_{i=0}^{N-2} A^{i}}} \sum_{a_{1} \in A \cup\{\dagger\}} \mu_{\mathrm{L}}\left(x, x a_{1} \right) \\ & =\# \mathrm{ob}(\mathrm{~L})-\#\left\{x \in \mathrm{ob}(\mathrm{~L}): x=\perp a \text { with } a \in \bigcup_{i=0}^{N-2} A^{i}\right\}(\# A+1) \end{aligned}	$\begin{aligned} \operatorname{Mag}(\mathrm{L}) & = \sum_{x, y \in \mathrm{ob}(\mathrm{~L})} \zeta_{ \mathrm{L}}^{-1}(x, y) \\ & =\sum_{x, y \in \mathrm{ob}(\mathrm{~L})} \mu_{\mathrm{L}}(x, y) \\ & =\sum_{x \in \mathrm{ob}(\mathrm{~L})} \mu_{\mathrm{L}}(x, x)+ \sum_{\substack{x=\perp a \in \mathrm{ob}(\mathrm{~L}) \\ a \in \bigcup_{i=0}^{N-2} A^{i}}} \sum_{a_{1} \in A \cup\{\dagger\}} \mu_{\mathrm{L}}\left(x, x a_{1} \right) \\ & =\# \mathrm{ob}(\mathrm{~L})-\#\left\{x \in \mathrm{ob}(\mathrm{~L}): x=\perp a \text { with } a \in \bigcup_{i=0}^{N-2} A^{i}\right\}(\# A+1) \end{aligned}$	$\begin{aligned} \operatorname{Mag}(\mathsf{L}) &= \sum_{x,y\in\operatorname{ob}(\mathsf{L})} \zeta_{\mathsf{L}}^{-1}(x,y) \\ &= \sum_{x,y\in\operatorname{ob}(\mathsf{L})} \mu_{\mathsf{L}}(x,y) \\ &= \sum_{x\in\operatorname{ob}(\mathsf{L})} \mu_{\mathsf{L}}(x,x) + \sum_{\substack{x=\perp a\in\operatorname{ob}(\mathsf{L})\\ a\in\bigcup_{i=0}^{N-2}A^i}} \sum_{a_1\in A\cup\{\dagger\}} \mu_{\mathsf{L}}(x,xa_1) \\ &= \#\operatorname{ob}(\mathsf{L}) - \#\left\{x\in\operatorname{ob}(\mathsf{L}) : x=\perp a \text{ with } a\in\bigcup_{i=0}^{N-2}A^i\right\}(\#A+1). \end{aligned}$
2501.06662_EQ0022	015	■ 0.871 ● N|+2	\begin{aligned} \# \mathrm{ob}(\mathrm{~L}) & =1+ \sum_{i=1}^{N-1} \# \mathrm{~L}^{(i)} \\ & =1+ \sum_{i=1}^{N-1} \# A^{i}+\# A^{i-1}=\# A^{N-1}+2\left(\# A^{N-2}\right)+\cdots+2(\# A)+2 \end{aligned}	$\begin{aligned} \# \mathrm{ob}(\mathrm{~L}) & =1+ \sum_{i=1}^{N-1} \# \mathrm{~L}^{(i)} \\ & =1+ \sum_{i=1}^{N-1} \# A^{i}+\# A^{i-1}=\# A^{N-1}+2\left(\# A^{N-2}\right)+\cdots+2(\# A)+2 \end{aligned}$	$\begin{aligned} \#\operatorname{ob}(\mathsf{L}) &= 1+\sum_{i=1}^{N-1}\#\mathsf{L}^{(i)} \\ &= 1+\sum_{i=1}^{N-1}\#A^i+\#A^{i-1} = \#A^{N-1}+2(\#A^{N-2})+\cdots+2(\#A)+2. \end{aligned}$
2501.06662_EQ0045	019	■ 0.922 ● N|+1	\zeta_{\mathcal{M}_{x}}^{-1}=\left.\zeta_{ \mathcal{M}}^{-1}\right|_{\operatorname{ob}\left(\mathcal{M}_{x}\right) \times \operatorname{ob}\left(\mathcal{M}_{x}\right)}	$\zeta_{\mathcal{M}_{x}}^{-1}=\left.\zeta_{ \mathcal{M}}^{-1}\right\|_{\operatorname{ob}\left(\mathcal{M}_{x}\right) \times \operatorname{ob}\left(\mathcal{M}_{x}\right)}$	$\zeta_{\mathcal{M}_x}^{-1} = \zeta_{\mathcal{M}}^{-1}\big\|_{\operatorname{ob}(\mathcal{M}_x)\times\operatorname{ob}(\mathcal{M}_x)},$
2501.06662_ EQ0039 (12)	018	■ 0.940 ● N|+0	f^{\prime}(1)=\sum_{x \in \mathrm{ob}(\mathcal{M}) \backslash T(\perp)} H\left(p_{x}\right) .	$f^{\prime}(1)=\sum_{x \in \mathrm{ob}(\mathcal{M}) \backslash T(\perp)} H\left(p_{x}\right) .$	$f'(1) = \sum_{x\in\operatorname{ob}(\mathcal{M})\setminus T(\perp)} H(p_x).$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

<table>
<tr><th>Identifier</th><th>Page</th><th>Conf.</th><th>LaTeX source</th><th>Rendered</th><th>Scan image</th></tr>
<tr><td>2501.06662_EQ0046</td><td>020</td><td>0.942
W|+2</td><td><code>\begin{aligned} \operatorname{Mag}\left(t \mathcal{M}_{x} \right) & =\sum_{y, z \in \mathrm{ob}\left(\mathcal{M}_{x}\right)} \zeta_{\mathcal{M}_{x}}^{-1}(y, z) \\ & =\sum_{y \in \mathrm{ob}\left(\mathcal{M}_{x} \right)} \zeta_{\mathcal{M}_{x}}^{-1}(y, y)-\sum_{y \in \mathrm{ob}\left(\mathcal{M}_{x}\right) \backslash T(x)} \sum_{y \in \mathrm{~L}_{y}^{(1)}} \pi(z \mid y)^{t} \ \ & =(t-1) \sum_{y \in \mathrm{ob}\left(\mathcal{M}_{x} \right) \backslash T(x)} H_{t}\left(p_{y}\right)+\#(T(x)) . \end{aligned}</code></td><td>$$\begin{aligned} \operatorname{Mag}\left(t \mathcal{M}_{x} \right) & =\sum_{y, z \in \mathrm{ob}\left(\mathcal{M}_{x}\right)} \zeta_{\mathcal{M}_{x}}^{-1}(y, z) \\ & =\sum_{y \in \mathrm{ob}\left(\mathcal{M}_{x} \right)} \zeta_{\mathcal{M}_{x}}^{-1}(y, y)-\sum_{y \in \mathrm{ob}\left(\mathcal{M}_{x}\right) \backslash T(x)} \sum_{y \in \mathrm{~L}_{y}^{(1)}} \pi(z \mid y)^{t} \\ & =(t-1) \sum_{y \in \mathrm{ob}\left(\mathcal{M}_{x} \right) \backslash T(x)} H_{t}\left(p_{y}\right)+\#(T(x)) . \end{aligned}$$</td><td>$$\begin{aligned} \operatorname{Mag}(t \mathcal{M}_{x}) & =\sum_{y, z \in \mathrm{ob}(\mathcal{M}_{x})} \zeta_{\mathcal{M}_{x}}^{-1}(y, z) \\ & =\sum_{y \in \mathrm{ob}(\mathcal{M}_{x})} \zeta_{\mathcal{M}_{x}}^{-1}(y, y)-\sum_{y \in \mathrm{ob}(\mathcal{M}_{x}) \backslash T(x)} \sum_{y \in \mathsf{L}_{y}^{(1)}} \pi(z|y)^{t} \\ & =(t-1) \sum_{y \in \mathrm{ob}(\mathcal{M}_{x}) \backslash T(x)} H_{t}(p_{y})+\#(T(x)). \end{aligned}$$</td></tr>
<tr><td>2501.06662_EQ0040</td><td>019</td><td>0.945
N|+0</td><td><code>f(t)=\#(\mathrm{ob}(\mathcal{M}))-\sum_{x \in \mathrm{ob}(\mathcal{M}) \backslash T(\perp)} Z_{x}(t),</code></td><td>$$f(t)=\#(\mathrm{ob}(\mathcal{M}))-\sum_{x \in \mathrm{ob}(\mathcal{M}) \backslash T(\perp)} Z_{x}(t),$$</td><td>$$f(t)=\#(\mathrm{ob}(\mathcal{M}))-\sum_{x \in \mathrm{ob}(\mathcal{M}) \backslash T(\perp)} Z_{x}(t),$$</td></tr>
<tr><td>2501.06662_EQ0044</td><td>019</td><td>0.971
N|+0</td><td><code>\zeta_{\mathcal{M}_{x}}=\left.\zeta_{\mathcal{M}} \right|_{\operatorname{ob}\left(\mathcal{M}_{x}\right) \times \operatorname{ob}\left(\mathcal{M}_{x}\right)}</code></td><td>$$\zeta_{\mathcal{M}_{x}}=\left.\zeta_{\mathcal{M}} \right|_{\operatorname{ob}\left(\mathcal{M}_{x}\right) \times \operatorname{ob}\left(\mathcal{M}_{x}\right)}$$</td><td>$$\zeta_{\mathcal{M}_{x}}=\left.\zeta_{\mathcal{M}} \right|_{\mathrm{ob}(\mathcal{M}_{x}) \times \mathrm{ob}(\mathcal{M}_{x})}$$</td></tr>
<tr><td>2501.06662_EQ0032</td><td>017</td><td>0.973
N|+0</td><td><code>\pi(y \mid x)^{t} \sum_{k \geq 0}(-1)^{k} \#\{\text { nondegenerate paths of length } k \text { from } x \text { to } y \text { in } \mathrm{L}\}</code></td><td>$$\pi(y \mid x)^{t} \sum_{k \geq 0}(-1)^{k} \#\{\text { nondegenerate paths of length } k \text { from } x \text { to } y \text { in } \mathrm{L}\}$$</td><td>$$\pi(y|x)^{t} \sum_{k \geq 0}(-1)^{k} \#\{\text{nondegenerate paths of length } k \text{ from } x \text{ to } y \text{ in } \mathsf{L}\}$$</td></tr>
<tr><td>2501.06662_EQ0056</td><td>022</td><td>0.986
N|+0</td><td><code>P P L(y)=1 / \zeta_{t}\left(a_{0}, y\right)=1 / e^{t \ln \pi\left(y \mid a_{0}\right)},</code></td><td>$$P P L(y)=1 / \zeta_{t}\left(a_{0}, y\right)=1 / e^{t \ln \pi\left(y \mid a_{0}\right)},$$</td><td>$$PPL(y)=1/\zeta_{t}(a_{0}, y)=1/e^{t \ln \pi(y|a_{0})},$$</td></tr>
<tr><td>2501.06662_ EQ0015 (7)</td><td>012</td><td>0.988
N|+0</td><td><code>\operatorname{Mag}(\mathcal{C})=\sum_{(x, y) \in \operatorname{ob}(\mathcal{C}) \times \operatorname{ob}(\mathcal{C})} \zeta_{\mathcal{C}}^{-1}(x, y) .</code></td><td>$$\operatorname{Mag}(\mathcal{C})=\sum_{(x, y) \in \operatorname{ob}(\mathcal{C}) \times \operatorname{ob}(\mathcal{C})} \zeta_{\mathcal{C}}^{-1}(x, y) .$$</td><td>$$\operatorname{Mag}(\mathcal{C})=\sum_{(x, y) \in \mathrm{ob}(\mathcal{C}) \times \mathrm{ob}(\mathcal{C})} \zeta_{\mathcal{C}}^{-1}(x, y).$$</td></tr>
<tr><td>2501.06662_EQ0002</td><td>004</td><td>0.991
N|+0</td><td><code>\frac{(t-1)\left(n^{1-t}-1\right)}{1-t} \cdot \#(\operatorname{ob}(\mathcal{M}) \backslash T(\perp))+ \#(T(\perp))</code></td><td>$$\frac{(t-1)\left(n^{1-t}-1\right)}{1-t} \cdot \#(\operatorname{ob}(\mathcal{M}) \backslash T(\perp))+ \#(T(\perp))$$</td><td>$$\frac{(t-1)(n^{1-t}-1)}{1-t} \cdot \#(\mathrm{ob}(\mathcal{M}) \setminus T(\perp))+\#(T(\perp))$$</td></tr>
<tr><td>2501.06662_EQ0005</td><td>006</td><td>0.991
N|+0</td><td><code>\operatorname{ob}(\mathrm{L})=\left\{\perp a: a \in A^{*} \text { and }|a| \leq N-1\right\} \sqcup\left\{\perp a \dagger: a \in A^{*} \text { and }|a|<N-1\right\} .</code></td><td>$$\operatorname{ob}(\mathrm{L})=\left\{\perp a: a \in A^{*} \text { and }|a| \leq N-1\right\} \sqcup\left\{\perp a \dagger: a \in A^{*} \text { and }|a|<N-1\right\} .$$</td><td>$$\mathrm{ob}(\mathsf{L})=\{\perp a: a \in A^{*} \text{ and } |a| \leq N-1\} \sqcup \{\perp a \dagger: a \in A^{*} \text{ and } |a|<N-1\}.$$</td></tr>
<tr><td>2501.06662_EQ0006</td><td>008</td><td>0.996
C|+26</td><td><code>\begin{aligned} & T(x)=\left\{y \in \operatorname{ob} \left(\mathrm{~L}_{x}\right): y \text { is an unfinished text of length } N\right. \text { or } \\ & \qquad y \text { is a finished text such that }|y| \leq N\} . \end{aligned}</code></td><td>$$\begin{aligned} & T(x)=\left\{y \in \operatorname{ob} \left(\mathrm{~L}_{x}\right): y \text { is an unfinished text of length } N\right. \text { or } \\ & \qquad y \text { is a finished text such that }|y| \leq N\} . \end{aligned}$$</td><td>$$\begin{aligned} T(x)=\{y \in \mathrm{ob}(\mathsf{L}_{x}): y \text{ is an unfinished text of length } N \text{ or} \\ y \text{ is a finished text such that } |y| \leq N\}. \end{aligned}$$</td></tr>
<tr><td colspan="6">Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value
Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured</td></tr>
</table>

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
2501.06662_EQ0034	017	0.998 N|+0	`\operatorname{Mag}(t \mathcal{M}):=\sum_{x, y \in \operatorname{ob}(\mathcal{M})} \zeta_{t}^{-1}(x, y) .`	$\operatorname{Mag}(t \mathcal{M}):=\sum_{x, y \in \operatorname{ob}(\mathcal{M})} \zeta_{t}^{-1}(x, y) .$	$\operatorname{Mag}(t\mathcal{M}) := \sum_{x,y\in\operatorname{ob}(\mathcal{M})} \zeta_t^{-1}(x,y).$
2501.06662_EQ0003	004	0.998 W|+0	`\underbrace{\#(T(\perp))}_{\text {deterministic LM }} \leq \lim _{t \rightarrow \infty} \operatorname{Mag}(t \mathcal{M}) \leq \underbrace{\#(\operatorname{ob}(\mathcal{M}))}_{\text {maximally random LM }} .`	$\underbrace{\#(T(\perp))}_{\text {deterministic LM }} \leq \lim _{t \rightarrow \infty} \operatorname{Mag}(t \mathcal{M}) \leq \underbrace{\#(\operatorname{ob}(\mathcal{M}))}_{\text {maximally random LM }} .$	$\underbrace{\#(T(\perp))}_{\text{deterministic LM}} \leq \lim_{t\to\infty} \operatorname{Mag}(t\mathcal{M}) \leq \underbrace{\#(\operatorname{ob}(\mathcal{M}))}_{\text{maximally random LM}} .$
2501.06662_ EQ0054 (14)	022	0.998 N|+0	`\operatorname{Mag}(t \mathcal{M})=\operatorname{rank} H_{0,0}(\mathcal{M})-\sum_{\ell \geq 0} q^{\ell} \operatorname{rank} H_{1, \ell}(\mathcal{M}) .`	$\operatorname{Mag}(t \mathcal{M})=\operatorname{rank} H_{0,0}(\mathcal{M})-\sum_{\ell \geq 0} q^{\ell} \operatorname{rank} H_{1, \ell}(\mathcal{M}) .$	$\operatorname{Mag}(t\mathcal{M}) = \operatorname{rank} H_{0,0}(\mathcal{M}) - \sum_{\ell\geq 0} q^\ell \operatorname{rank} H_{1,\ell}(\mathcal{M}).$
2501.06662_EQ0019	014	0.999 N|+0	`\mu_{P}(s, u)= \begin{cases}1 & \text { if } s=u \\ -\sum_{s \leq t<u} \mu_{P}(s, t) & \text { if } s<u \\ 0 & \text { otherwise } .\end{cases}`	$\mu_{P}(s, u)= \begin{cases}1 & \text { if } s=u \\ -\sum_{s \leq t<u} \mu_{P}(s, t) & \text { if } s<u \\ 0 & \text { otherwise } .\end{cases}$	$\mu_P(s,u) = \begin{cases} 1 & \text{if } s=u \\ -\sum\limits_{s\leq t<u} \mu_P(s,t) & \text{if } s<u \\ 0 & \text{otherwise.} \end{cases}$
2501.06662_EQ0042	019	0.999 N|+0	`H\left(p_{x}\right)=-\left.\frac{\mathrm{d}}{\mathrm{~d} t} Z_{x}(t)\right|_{t=1},`	$H\left(p_{x}\right)=-\left.\frac{\mathrm{d}}{\mathrm{~d} t} Z_{x}(t)\right\vert_{t=1},$	$H(p_x) = -\left.\frac{\mathrm{d}}{\mathrm{d}t} Z_x(t)\right\vert_{t=1},$
2501.06662_EQ0004	005	1.000 N|+0	`\operatorname{Mag}(t \mathcal{M})=\sum_{\ell} e^{-t \ell} \sum_{k \geq 0}(-1)^{k} \operatorname{rank}\left(H_{k, \ell}(\mathcal{M})\right),`	$\operatorname{Mag}(t \mathcal{M})=\sum_{\ell} e^{-t \ell} \sum_{k \geq 0}(-1)^{k} \operatorname{rank}\left(H_{k, \ell}(\mathcal{M})\right),$	$\operatorname{Mag}(t\mathcal{M}) = \sum_\ell e^{-t\ell} \sum_{k\geq 0} (-1)^k \operatorname{rank}(H_{k,\ell}(\mathcal{M})),$
2501.06662_EQ0012	011	1.000 N|+1	`\begin{aligned} & x=\perp a_{1} \cdots a_{t} \\ & y=\perp a_{1} \cdots a_{t} \cdots a_{t+k} \\ & z=\perp a_{1} \cdots a_{t} \cdots a_{t+k} \cdots a_{t+k+k^{\prime}} \end{aligned}`	$\begin{aligned} & x=\perp a_{1} \cdots a_{t} \\ & y=\perp a_{1} \cdots a_{t} \cdots a_{t+k} \\ & z=\perp a_{1} \cdots a_{t} \cdots a_{t+k} \cdots a_{t+k+k^{\prime}} \end{aligned}$	$\begin{aligned} x &= \perp a_1 \cdots a_t \\ y &= \perp a_1 \cdots a_t \cdots a_{t+k} \\ z &= \perp a_1 \cdots a_t \cdots a_{t+k} \cdots a_{t+k+k'}. \end{aligned}$
2501.06662_EQ0031	017	1.000 N|+1	`\begin{aligned} \prod_{i=1}^{k} \pi\left(y_{i} \mid y_{i-1}\right)^{t} & =\pi\left(y_{1} \mid y_{0} \right)^{t} \pi\left(y_{2} \mid y_{1}\right)^{t} \pi \left(y_{3} \mid y_{2}\right)^{t} \cdots \pi\left(y_{k} \mid y_{k-1}\right)^{t} \\ & =\pi\left(y_{2} \mid y_{0} \right)^{t} \pi\left(y_{3} \mid y_{2}\right)^{t} \cdots \pi\left(y_{k} \mid y_{k-1}\right)^{t} \\ & =\quad \vdots \\ & =\pi\left(y_{k} \mid y_{0}\right)^{t} \end{aligned}`	$\begin{aligned} \prod_{i=1}^{k} \pi\left(y_{i} \mid y_{i-1}\right)^{t} & =\pi\left(y_{1} \mid y_{0} \right)^{t} \pi\left(y_{2} \mid y_{1}\right)^{t} \pi \left(y_{3} \mid y_{2}\right)^{t} \cdots \pi\left(y_{k} \mid y_{k-1}\right)^{t} \\ & =\pi\left(y_{2} \mid y_{0} \right)^{t} \pi\left(y_{3} \mid y_{2}\right)^{t} \cdots \pi\left(y_{k} \mid y_{k-1}\right)^{t} \\ & =\quad \vdots \\ & =\pi\left(y_{k} \mid y_{0}\right)^{t} \end{aligned}$	$\begin{aligned} \prod_{i=1}^k \pi(y_i\vert y_{i-1})^t &= \pi(y_1\vert y_0)^t \pi(y_2\vert y_1)^t \pi(y_3\vert y_2)^t \cdots \pi(y_k\vert y_{k-1})^t \\ &= \pi(y_2\vert y_0)^t \pi(y_3\vert y_2)^t \cdots \pi(y_k\vert y_{k-1})^t \\ &= \quad \vdots \\ &= \pi(y_k\vert y_0)^t. \end{aligned}$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
2501.06662_EQ0007	008	1.000 K|+0	`p\left(a_{3} \mid \perp a\right) p\left(a_{4} \mid \perp a a_{3}\right) p\left(a_{5} \mid \perp a a_{3} a_{4} \right) p\left(\dagger \mid \perp a a_{3} a_{4} a_{5} \right) .`	$p\left(a_{3} \mid \perp a\right) p\left(a_{4} \mid \perp a a_{3}\right) p\left(a_{5} \mid \perp a a_{3} a_{4} \right) p\left(\dagger \mid \perp a a_{3} a_{4} a_{5} \right) .$	$p(a_3 \mid \perp a)p(a_4 \mid \perp aa_3)p(a_5 \mid \perp aa_3a_4)p(\dagger \mid \perp aa_3a_4a_5).$
2501.06662_EQ0008 (1)	009	1.000 W|+0	`\pi(y \mid x):=\left\{\begin{array}{ll} 1 & \text { if } x=y \\ 0 & \text { if } x \nrightarrow y \\ \prod_{i=1}^{k} p\left(a_{t+i} \mid y_{<t+i}\right) & \text { if } x \rightarrow y \end{array} .\right.`	$\pi(y \mid x):=\left\{\begin{array}{ll} 1 & \text { if } x=y \\ 0 & \text { if } x \nrightarrow y \\ \prod_{i=1}^{k} p\left(a_{t+i} \mid y_{<t+i}\right) & \text { if } x \rightarrow y \end{array} .\right.$	$\pi(y \mid x) := \begin{cases} 1 & \text{if } x = y \\ 0 & \text{if } x \nrightarrow y \\ \prod_{i=1}^{k} p(a_{t+i} \mid y_{<t+i}) & \text{if } x \rightarrow y \end{cases}.$
2501.06662_EQ0009 (2)	009	1.000 N|+0	`\sum_{y \in T(x)} \pi(y \mid x)=\sum_{a \in A \cup\{ \dagger\}} p_{x}(a)=1,`	$\sum_{y \in T(x)} \pi(y \mid x)=\sum_{a \in A \cup\{ \dagger\}} p_{x}(a)=1,$	$\sum_{y \in T(x)} \pi(y \mid x) = \sum_{a \in A \cup \{\dagger\}} p_x(a) = 1,$
2501.06662_EQ0010 (4)	009	1.000 W|+0	`\begin{aligned} \sum_{y \in T(x)} \pi(y \mid x) & = \sum_{i=0}^{m-1} \sum_{a^{\prime} \in A^{i}} \pi\left(x a^{\prime} \dagger \mid x\right)+\sum_{a \in A^{m}} \pi(x a \mid x) \\ & =\sum_{i=0}^{m-1} \sum_{a^{\prime} \in A^{i}} \pi\left(x a^{\prime} \dagger \mid x\right)+ \sum_{\substack{a=a^{\prime} a^{\prime \prime} \\ a^{ \prime} \in A^{m-1}, a^{\prime \prime} \in A}} p\left(a^{ \prime \prime} \mid x a^{\prime}\right) \pi\left(x a^{ \prime} \mid x\right) \\ & =\sum_{i=0}^{m-2} \sum_{a^{ \prime} \in A^{i}} \pi\left(x a^{\prime} \dagger \mid x\right)+\sum_{a^{\prime} \in A^{m-1}} \pi\left(x a^{ \prime} \mid x\right) \sum_{a^{\prime \prime} \in A \cup\{\dagger\}} p\left(a^{\prime \prime} \mid x a^{ \prime}\right) \\ & =\sum_{i=0}^{m-2} \sum_{a^{\prime} \in A^{i}} \pi\left(x a^{\prime} \dagger \mid x\right)+ \sum_{a^{\prime} \in A^{m-1}} \pi\left(x a^{\prime} \mid x\right) \end{aligned}`	$\begin{aligned} \sum_{y \in T(x)} \pi(y \mid x) & = \sum_{i=0}^{m-1} \sum_{a^{\prime} \in A^{i}} \pi\left(x a^{\prime} \dagger \mid x\right)+\sum_{a \in A^{m}} \pi(x a \mid x) \\ & =\sum_{i=0}^{m-1} \sum_{a^{\prime} \in A^{i}} \pi\left(x a^{\prime} \dagger \mid x\right)+ \sum_{\substack{a=a^{\prime} a^{\prime \prime} \\ a^{ \prime} \in A^{m-1}, a^{\prime \prime} \in A}} p\left(a^{ \prime \prime} \mid x a^{\prime}\right) \pi\left(x a^{ \prime} \mid x\right) \\ & =\sum_{i=0}^{m-2} \sum_{a^{ \prime} \in A^{i}} \pi\left(x a^{\prime} \dagger \mid x\right)+\sum_{a^{\prime} \in A^{m-1}} \pi\left(x a^{ \prime} \mid x\right) \sum_{a^{\prime \prime} \in A \cup\{\dagger\}} p\left(a^{\prime \prime} \mid x a^{ \prime}\right) \\ & =\sum_{i=0}^{m-2} \sum_{a^{\prime} \in A^{i}} \pi\left(x a^{\prime} \dagger \mid x\right)+ \sum_{a^{\prime} \in A^{m-1}} \pi\left(x a^{\prime} \mid x\right) \end{aligned}$	$\begin{aligned} \sum_{y \in T(x)} \pi(y \mid x) &= \sum_{i=0}^{m-1} \sum_{a' \in A^i} \pi(xa' \dagger \mid x) + \sum_{a \in A^m} \pi(xa \mid x) \\ &= \sum_{i=0}^{m-1} \sum_{a' \in A^i} \pi(xa' \dagger \mid x) + \sum_{\substack{a=a'a'' \\ a' \in A^{m-1}, a'' \in A}} p(a'' \mid xa')\pi(xa' \mid x) \\ &= \sum_{i=0}^{m-2} \sum_{a' \in A^i} \pi(xa' \dagger \mid x) + \sum_{a' \in A^{m-1}} \pi(xa' \mid x) \sum_{a'' \in A \cup \{\dagger\}} p(a'' \mid xa') \\ &= \sum_{i=0}^{m-2} \sum_{a' \in A^i} \pi(xa' \dagger \mid x) + \sum_{a' \in A^{m-1}} \pi(xa' \mid x) \end{aligned}$
2501.06662_EQ0011	010	1.000 N|+0	`\begin{aligned} 1 & \leq \mathcal{C}(x, x) \\ \mathcal{C}(y, z) \otimes \mathcal{C}(x, y) & \leq \mathcal{C}(x, z) \end{aligned}`	$\begin{aligned} 1 & \leq \mathcal{C}(x, x) \\ \mathcal{C}(y, z) \otimes \mathcal{C}(x, y) & \leq \mathcal{C}(x, z) \end{aligned}$	$\begin{aligned} 1 &\leq \mathcal{C}(x, x) \\ \mathcal{C}(y, z) \otimes \mathcal{C}(x, y) &\leq \mathcal{C}(x, z) \end{aligned}$

Conf. MathPix confidence: ≥0.9 0.5–0.9 <0.5 — no value

Residual render vs scan (inkdrill): C component W weak S stable N noise K clean not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
2501.06662_ EQ0013 (6)	011	■ 1.000 ● N|+2	`\begin{aligned} \pi(y \mid x) \pi(z \mid y) & = \prod_{i=1}^{k} p\left(a_{t+i} \mid y_{<t+i}\right) \prod_{j=1}^{k^{\prime}} p\left(a_{t+k+j} \mid z_{<t+k+j} \right) \\ & =\prod_{i=1}^{k+k^{\prime}} p\left(a_{t+i} \mid z_{<t+i}\right) \\ & =\pi(z \mid x) \end{aligned}`	$\begin{aligned} \pi(y \mid x) \pi(z \mid y) & = \prod_{i=1}^{k} p\left(a_{t+i} \mid y_{<t+i}\right) \prod_{j=1}^{k^{\prime}} p\left(a_{t+k+j} \mid z_{<t+k+j} \right) \\ & =\prod_{i=1}^{k+k^{\prime}} p\left(a_{t+i} \mid z_{<t+i}\right) \\ & =\pi(z \mid x) \end{aligned}$	$\begin{aligned} \pi(y\vert x)\pi(z\vert y) & = \prod_{i=1}^{k} p(a_{t+i}\vert y_{<t+i}) \prod_{j=1}^{k'} p(a_{t+k+j}\vert z_{<t+k+j}) \\ & = \prod_{i=1}^{k+k'} p(a_{t+i}\vert z_{<t+i}) \\ & = \pi(z\vert x). \end{aligned}$
2501.06662_EQ0014	012	■ 1.000 ● N|+0	`d(x, y):=-\ln \pi(y \mid x) .`	$d(x, y):=-\ln \pi(y \mid x) .$	$d(x,y) := -\ln \pi(y\vert x).$
2501.06662_EQ0016	013	■ 1.000 ● N|+0	`(f * g)(s, u):=\sum_{s \leq t \leq u} f(s, t) g(t, u),`	$(f * g)(s, u):=\sum_{s \leq t \leq u} f(s, t) g(t, u),$	$(f * g)(s,u) := \sum_{s\leq t\leq u} f(s,t)g(t,u),$
2501.06662_EQ0017	013	■ 1.000 ● K|+0	`\delta(s, u)= \begin{cases}1 & \text { if } s=u \\ 0 & \text { if } s \neq u\end{cases}`	$\delta(s, u)= \begin{cases}1 & \text { if } s=u \\ 0 & \text { if } s \neq u\end{cases}$	$\delta(s,u) = \begin{cases}1 & \text{if } s=u \\ 0 & \text{if } s\neq u\end{cases}$
2501.06662_EQ0018	013	■ 1.000 ● N|+0	`\zeta_{P}(s, u)=1, \quad \text { for all } s \leq u \text { in } P .`	$\zeta_{P}(s, u)=1, \quad \text { for all } s \leq u \text { in } P .$	$\zeta_P(s,u) = 1, \quad \text{for all } s\leq u \text{ in } P.$
2501.06662_EQ0020	014	■ 1.000 ● N|+0	`\begin{aligned} \mu_{\mathrm{L}}(x, x) & =1 \\ \mu_{\mathrm{L}}\left(x, x a_{1}\right) & =-\mu_{ \mathrm{L}}(x, x)=-1 \\ \mu_{\mathrm{L}}\left(x, x a_{1} a_{2}\right) & =-\mu_{\mathrm{L}}(x, x)-\mu_{ \mathrm{L}}\left(x, x a_{1}\right)=-1+1=0 \\ \mu_{ \mathrm{L}}\left(x, x a_{1} a_{2} a_{3}\right) & =- \mu_{\mathrm{L}}(x, x)-\mu_{\mathrm{L}}\left(x, x a_{1}\right)-\mu_{\mathrm{L}}\left(x, x a_{1} a_{2} \right)=-1+1+0=0 . \end{aligned}`	$\begin{aligned} \mu_{\mathrm{L}}(x, x) & =1 \\ \mu_{\mathrm{L}}\left(x, x a_{1}\right) & =-\mu_{ \mathrm{L}}(x, x)=-1 \\ \mu_{\mathrm{L}}\left(x, x a_{1} a_{2}\right) & =-\mu_{\mathrm{L}}(x, x)-\mu_{ \mathrm{L}}\left(x, x a_{1}\right)=-1+1=0 \\ \mu_{ \mathrm{L}}\left(x, x a_{1} a_{2} a_{3}\right) & =- \mu_{\mathrm{L}}(x, x)-\mu_{\mathrm{L}}\left(x, x a_{1}\right)-\mu_{\mathrm{L}}\left(x, x a_{1} a_{2} \right)=-1+1+0=0 . \end{aligned}$	$\begin{aligned} \mu_{\mathsf{L}}(x,x) & = 1 \\ \mu_{\mathsf{L}}(x,xa_1) & = -\mu_{\mathsf{L}}(x,x) = -1 \\ \mu_{\mathsf{L}}(x,xa_1a_2) & = -\mu_{\mathsf{L}}(x,x) - \mu_{\mathsf{L}}(x,xa_1) = -1+1 = 0 \\ \mu_{\mathsf{L}}(x,xa_1a_2a_3) & = -\mu_{\mathsf{L}}(x,x) - \mu_{\mathsf{L}}(x,xa_1) - \mu_{\mathsf{L}}(x,xa_1a_2) = -1+1+0 = 0. \end{aligned}$
2501.06662_EQ0024	015	■ 1.000 ● N|+1	`\zeta_{\mathcal{M}}(x, y):=e^{-d(x, y)}=\pi(y \mid x),`	$\zeta_{\mathcal{M}}(x, y):=e^{-d(x, y)}=\pi(y \mid x),$	$\zeta_{\mathcal{M}}(x,y) := e^{-d(x,y)} = \pi(y\vert x),$
2501.06662_EQ0025	015	■ 1.000 ● N|+0	`\left(\zeta_{\mathcal{M}}\right)_{t}(x, y):=e^{-t d(x, y)}=\pi(y \mid x)^{t} .`	$\left(\zeta_{\mathcal{M}}\right)_{t}(x, y):=e^{-t d(x, y)}=\pi(y \mid x)^{t} .$	$(\zeta_{\mathcal{M}})_t(x,y) := e^{-td(x,y)} = \pi(y\vert x)^t.$
2501.06662_ EQ0027 (9)	016	■ 1.000 ● N|+1	`\zeta_{t}^{-1}=\sum_{k \geq 0}(-1)^{k}\left(\zeta_{t}-\delta\right)^{k}`	$\zeta_{t}^{-1}=\sum_{k \geq 0}(-1)^{k}\left(\zeta_{t}-\delta\right)^{k}$	$\zeta_t^{-1} = \sum_{k\geq 0}(-1)^k(\zeta_t-\delta)^k,$
2501.06662_EQ0028	016	■ 1.000 ● N|+0	`\left(\zeta_{t}-\delta\right)(x, y)=\pi(x, y)^{t} \mathbf{1}_{(x, y) \in E},`	$\left(\zeta_{t}-\delta\right)(x, y)=\pi(x, y)^{t} \mathbf{1}_{(x, y) \in E},$	$(\zeta_t-\delta)(x,y) = \pi(x,y)^t\mathbf{1}_{(x,y)\in E},$
2501.06662_EQ0030	016	■ 1.000 ● N|+0	`\zeta_{t}^{-1}(x, y)=\pi(y \mid x)^{t} \zeta_{ \mathrm{L}}^{-1}(x, y) .`	$\zeta_{t}^{-1}(x, y)=\pi(y \mid x)^{t} \zeta_{ \mathrm{L}}^{-1}(x, y) .$	$\zeta_t^{-1}(x,y) = \pi(y\vert x)^t\,\zeta_{\mathsf{L}}^{-1}(x,y).$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
2501.06662_EQ0035	017	■ 1.000 ● K|+0	`H_{t}(p)=\frac{1}{t-1}\left(1-\sum_{i=1}^{n} p_{i}^{t}\right) .`	$H_{t}(p)=\frac{1}{t-1}\left(1-\sum_{i=1}^{n} p_{i}^{t}\right) .$	$H_t(p)=\frac{1}{t-1}\left(1-\sum_{i=1}^{n} p_i^t\right).$
2501.06662_EQ0038	018	■ 1.000 ● N|+0	`\lim _{t \rightarrow 1} H_{t}(p)=\lim _{t \rightarrow 1} \frac{1}{t-1}\left(1-\sum_{s \in S} p(s)^{t}\right)=-\sum_{s \in S} p(s) \ln p(s)=: H(p) .`	$\lim _{t \rightarrow 1} H_{t}(p)=\lim _{t \rightarrow 1} \frac{1}{t-1}\left(1-\sum_{s \in S} p(s)^{t}\right)=-\sum_{s \in S} p(s) \ln p(s)=: H(p) .$	$\lim_{t\rightarrow 1} H_t(p)=\lim_{t\rightarrow 1}\frac{1}{t-1}\left(1-\sum_{s\in S} p(s)^t\right)=-\sum_{s\in S} p(s)\ln p(s)=: H(p).$
2501.06662_EQ0041	019	■ 1.000 ● N|+0	`Z_{x}(t)=\sum_{a \in A \cup\{\dagger\}} p_{x}(a)^{t}=\sum_{a \in A \cup\{\dagger\}} e^{-t\left(-\ln p_{x}(a)\right)}`	$Z_{x}(t)=\sum_{a \in A \cup\{\dagger\}} p_{x}(a)^{t}=\sum_{a \in A \cup\{\dagger\}} e^{-t\left(-\ln p_{x}(a)\right)}$	$Z_x(t)=\sum_{a\in A\cup\{\dagger\}} p_x(a)^t=\sum_{a\in A\cup\{\dagger\}} e^{-t(-\ln p_x(a))}$
2501.06662_EQ0043	019	■ 1.000 ● N|+0	`\mathbb{E}_{\rho}(E)=\sum_{s \in S} E(s) \frac{e^{-t E(s)}}{\tilde{Z}(t)}=-\frac{\mathrm{d}}{\mathrm{~d} t} \ln \tilde{Z}(t)=\frac{f^{\prime}(t)}{\tilde{Z}(t)},`	$\mathbb{E}_{\rho}(E)=\sum_{s \in S} E(s) \frac{e^{-t E(s)}}{\tilde{Z}(t)}=-\frac{\mathrm{d}}{\mathrm{~d} t} \ln \tilde{Z}(t)=\frac{f^{\prime}(t)}{\tilde{Z}(t)},$	$\mathbb{E}_\rho(E)=\sum_{s\in S} E(s)\frac{e^{-tE(s)}}{\tilde{Z}(t)}=-\frac{\mathrm{d}}{\mathrm{d}t}\ln\tilde{Z}(t)=\frac{f'(t)}{\tilde{Z}(t)},$
2501.06662_EQ0047	020	■ 1.000 ● N|+0	`G_{k, \ell}=\left\{\left(y_{0}, \ldots, y_{k}\right) \in M^{k+1} \mid \sum_{i=0}^{k-1} d\left(y_{i}, y_{i+1}\right)=\ell \text { and for all } i, y_{i} \neq y_{i+1}\right\} .`	$G_{k, \ell}=\left\{\left(y_{0}, \ldots, y_{k}\right) \in M^{k+1} \mid \sum_{i=0}^{k-1} d\left(y_{i}, y_{i+1}\right)=\ell \text { and for all } i, y_{i} \neq y_{i+1}\right\} .$	$G_{k,\ell}=\left\{(y_0,\ldots,y_k)\in M^{k+1} \mid \sum_{i=0}^{k-1} d(y_i,y_{i+1})=\ell \text{ and for all } i,\, y_i\neq y_{i+1}\right\}.$
2501.06662_EQ0048	020	■ 1.000 ● K|+0	`\partial_{k}=\sum_{i=0}^{k}(-1)^{i} \partial_{k}^{i}`	$\partial_{k}=\sum_{i=0}^{k}(-1)^{i} \partial_{k}^{i}$	$\partial_k=\sum_{i=0}^{k}(-1)^i\partial_k^i$
2501.06662_EQ0050	021	■ 1.000 ● N|+0	`\operatorname{Mag}(t \mathcal{M})=\sum_{\ell} q^{\ell} \sum_{k \geq 0}(-1)^{k} \operatorname{rank}\left(H_{k, \ell}(\mathcal{M})\right)`	$\operatorname{Mag}(t \mathcal{M})=\sum_{\ell} q^{\ell} \sum_{k \geq 0}(-1)^{k} \operatorname{rank}\left(H_{k, \ell}(\mathcal{M})\right)$	$\operatorname{Mag}(t\mathcal{M})=\sum_{\ell} q^\ell\sum_{k\geq 0}(-1)^k\operatorname{rank}(H_{k,\ell}(\mathcal{M}))$
2501.06662_EQ0053	021	■ 1.000 ● N|+0	`\sum_{c \in G_{k, \ell}}(-1)^{k} e^{-t \ell}=(-1)^{k} q^{\ell} \# G_{k, \ell}=(-1)^{k} q^{\ell} \operatorname{rank}\left(M C_{k, \ell}(\mathcal{M})\right) .`	$\sum_{c \in G_{k, \ell}}(-1)^{k} e^{-t \ell}=(-1)^{k} q^{\ell} \# G_{k, \ell}=(-1)^{k} q^{\ell} \operatorname{rank}\left(M C_{k, \ell}(\mathcal{M})\right) .$	$\sum_{c\in G_{k,\ell}}(-1)^k e^{-t\ell}=(-1)^k q^\ell \# G_{k,\ell}=(-1)^k q^\ell\operatorname{rank}(MC_{k,\ell}(\mathcal{M})).$
2501.06662_EQ0055	022	■ 1.000 ● N|+0	`P P L(x)=\exp \left\{-\frac{1}{n} \sum_{i=1}^{n} \ln p\left(a_{i} \mid y_{<i}\right)\right\}`	$P P L(x)=\exp \left\{-\frac{1}{n} \sum_{i=1}^{n} \ln p\left(a_{i} \mid y_{<i}\right)\right\}$	$PPL(x)=\exp\left\{-\frac{1}{n}\sum_{i=1}^{n}\ln p(a_i\|y_{<i})\right\}$
2501.06662_EQ0057	023	■ 1.000 ● N|+0	`\theta(a, b)=0 \quad \text { if } \quad \mathrm{A}(a, b)=\varnothing .`	$\theta(a, b)=0 \quad \text { if } \quad \mathrm{A}(a, b)=\varnothing .$	$\theta(a,b)=0 \quad \text{if} \quad \mathsf{A}(a,b)=\varnothing.$
2501.06662_EQ0060	023	■ 1.000 ● N|+1	`-\sum_{a \in A} \sum_{a^{\prime} \in A} p_{\perp}(a) P\left(a, a^{\prime}\right) \log P\left(a, a^{\prime}\right),`	$-\sum_{a \in A} \sum_{a^{\prime} \in A} p_{\perp}(a) P\left(a, a^{\prime}\right) \log P\left(a, a^{\prime}\right),$	$-\sum_{a\in A}\sum_{a'\in A} p_\perp(a)P(a,a')\log P(a,a'),$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value
Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured